From Lagging to Leading: The Measured Emergence of Standing Capacitance Behind Consumer Connections, 1997–2025

D. Hume

***Abstract*—Power systems are transitioning from net-absorbing (lagging) to net-generating (leading) reactive power at light load; the ENTSO-E Final Reports on the 2025 Iberian and North Macedonian blackouts identify excess capacitive reactive power among the causal factors. Using regulatory half-hourly metering at New Zealand grid exit points, we measure the transition end to end over 29 years. On a balanced panel of 123 distribution grid exit points (the same sites in every year), mean overnight reactive power fell from about $+466$ MVAr (lagging) in 1997 to about -451 MVAr (leading) in 2025 (about -31 MVAr/yr), crossing net-leading overnight in 2016—measured, not a metering artefact, with its intensity and reach unchanged when every contaminated grid exit point is removed. A first-principles charging calculation with no fitted cable coefficient puts the demand-side share of the drift at about 80% (physical band 75–88%) over 2013–2025; two statistical methods corroborate the direction. The deepening is as large at the flat evening peak as overnight—the signature of standing shunt capacitance, not changed operating draw—and it is proportional to connections served: about 220 VAr per connection added over 2013–2025, crossing zero about 2012 and accumulating faster over the second half of that window than the first. The connected load fleet is adding standing distributed capacitance—consistent with the mandate-driven electromagnetic-compatibility filter capacitance in practically every switched-mode device (a labelled hypothesis); both of the scaling test's inputs are records any operator holds. The reversal relocates the binding voltage-security question to the overvoltage direction, where the defences, margins, and load models of practice are thinner or addressed to the opposite direction. We set out what this implies for the load models, security margins, and emergency schemes of power-system practice.**

***Index Terms*—Reactive power, power factor, overvoltage, leading power factor, shunt capacitance, distribution cabling, measurement, grid exit point, load composition, load modelling, power-system analysis.**

I. Introduction

POWER systems internationally are shifting from net-absorbing (lagging) toward net-generating (leading) reactive power at light load, as the connected load fleet modernises toward power-electronic devices and networks add underground cabling. The direction of the transition is not in dispute, and this paper does not claim to be the first to observe it. What has been missing is, first, a long *measured* record of the transition observed end to end at supply-point resolution, and second, a measured separation of how much of the drift is the load fleet and how much is the network. This paper supplies both for a single national system, connects the result to a failure mode that has recently moved from theory to realised blackouts, and closes with what the reversal implies for the analysis and defences that were built for the opposite regime. The trail ends somewhere unexpected: at the electromagnetic-compatibility filter inside practically every mains-connected device—capacitance that one part of the regulatory apparatus drives in and no other part counts—advanced here as a labelled hypothesis with its decisive test identified.

A. A global reactive transition, already observed

The phenomenon is well attested across jurisdictions: a steep decline in reactive demand at minimum load in Great Britain, with distribution networks beginning to export reactive power to the transmission system [1], [2]; off-peak overvoltage episodes attributed by the French transmission operator to distributed generation, underground cabling, and evolving final consumption [3]; a rising likelihood of overvoltage events linked partly to more energy-efficient appliances in Queensland [4]; rising capacitive export to the extra-high-voltage network in continental Europe [5]; and, at device level, a majority of modern household appliances presenting a leading power factor [6], with expert reviews documenting the error constant-power-factor load modelling introduces [7], [8]. No consumer-device standard requires a unity fundamental-frequency (displacement) power factor: IEC 61000-3-2 limits harmonic current only [9], and the one device-level power-factor family in force anywhere (the European lighting floors [10]) is a full-load, direction-blind metric that never sees a device's standing draw; the extended version [11] carries the fuller instrument survey. The overvoltage-instability literature itself now records the observable: the 2026 review of overvoltage instability mechanisms notes, among its causes of voltage rise, that cable substitution has "change[d] the power factor to leading in several urban substations" [12]—an observation offered without citation, and attributed wholly to the network's cables rather than to any change in what the connected demand draws. Every system in this literature runs 230-volt mains; Section V-B returns to that regularity.

The closest measured system-level series is that British one: roughly eight years (about 2005–2013) of reactive de-

D. Hume is with the Electricity Authority, Wellington, New Zealand (e-mail: corresponding author details on submission). ORCID iD: https://orcid.org/0009-0006-6920-0598.

This is descriptive analysis of public regulatory metering and information-disclosure data. The views expressed are the author's. No external funding was received. The analysis informs, but does not advocate, a live reactive-power pricing policy question (Section VI-D).

A reproducibility package (the analysis notebooks that compute every number and figure in this paper from public data) accompanies the work.

mand at grid supply points, projected forward from load-composition modelling and scenario projection—neither a multi-decade measured record nor a quantified demand-versus-network share of the realised drift; boundary studies in the Netherlands [13], Denmark [14], and Slovakia [5] likewise characterise or simulate the export without either. The study closest to the present decomposition is Bosisio *et al.* [15] on the Milan distribution network: measured 15-minute boundary data at 37 high-voltage/medium-voltage transformers, naming the same three-term balance used here and, in the indexed English-language literature to our knowledge, the only prior study to articulate this split explicitly on measured boundary data. It compares two years across a COVID-19 demand shock, assigns no numerical share, and reads the dominant term as the cables, opposite to ours; we treat it as complementary (it confirms the framework on measured data for one heavily cabled city, and its cable-dominant reading is consistent with cable intensity shaping where net-leading territory is reached first, Section IV-A), and it leaves open the national, multi-decade, quantified attribution (the fuller positioning is in the extended version [11]).

B. The realised risk

In 2025 two power systems suffered major blackouts whose official investigations found a leading-reactive/overvoltage mechanism at their core: the Final Reports of the European Network of Transmission System Operators for Electricity (ENTSO-E), both published 2026, on the Iberian Peninsula event of 28 April 2025 [16] and on the North Macedonia event of 18 May 2025 [17], detailed in Section VI-C. What matters here is what those investigations could not do. Both reports treat the capacitive surplus as a network-side quantity and low demand as a magnitude rather than as a fleet whose reactive character has itself changed; neither distinguishes nor quantifies a demand-side composition shift from network charging; and neither report states the aggregate reactive character of demand anywhere (the extended version [11] carries the reports' load-model detail). These are not criticisms of investigations that reasoned from the data available to them; they identify the space a long measured record can fill. The field's current synthesis has the same shape: the 2026 mechanisms review [12] gives as its causes of voltage rise corridors below surge-impedance loading, cable substitution, synchronous-generator decommissioning, and fixed-power-factor converter operation—network and generation throughout—representing load as lagging or unity power factor in every model, citing no measurement of demand reactive character, and calling for none.

C. Contribution

This paper makes a layered contribution.

Contribution 1 (measured trajectory). Using regulatory half-hourly active and reactive power at New Zealand grid exit points, we measure the system's reactive character at supply-point resolution over 29 years (1997–2025) on a balanced panel—the same grid exit points in every year, so sites entering or leaving the record cannot manufacture the trend—characterise its rate and crossing, and show the trajectory is a measured phenomenon and not a metering artefact. The computation reuses settlement metering that any operator holds, so the same measurement is available to any system.

Contribution 2 (decomposition of the drift). We decompose the drift into a demand-side load-composition component and a network cable-charging component. The magnitude is carried by a first-principles charging calculation that computes each network's cable charging directly from its disclosed circuit kilometres (Section II-D) and so fits no cable coefficient; evaluated over the 2013–2025 disclosure window, it puts the demand-side (*organic*) share of the drift at about 80% (physical band 75–88%), with two statistical methods corroborating the direction over the full 29-year record but not themselves pinning the share. We report a measured direction and dominance and do not claim a point-identified split.

Contribution 3 (the term identified). We then identify what the demand-side driver is physically doing. The deepening is load-independent—as large at the flat evening peak as overnight—so it behaves as standing shunt capacitance, permanently energised, not as a change in how the load draws reactive power while operating. And it lives behind the connections: across grid exit points the 2013–2025 deepening is proportional to the number of connections served, with no detectable connection-independent residual; refitted year by year, each connection's standing contribution sits slightly on the inductive side of zero at the start of the record, crosses zero in 2012, and grows every year thereafter—about 220 VAr per connection added over 2013–2025, the same figure whether fitted to the endpoint change or read off the level history—and it accumulated faster over the second half of that window than the first (Section V). The load fleet is accumulating standing distributed capacitance, not merely drawing fewer lagging vars. We advance the mandated electromagnetic-compatibility filter capacitance as its candidate device layer (a labelled hypothesis), identify the decisive low-voltage measurement that would settle it, and note that both of the scaling test's inputs—settlement metering and connection counts—are records any operator holds.

The significance is what the reversal does to analysis: the direction of the binding voltage-security question reverses; the automatic last-resort defences and the margin tooling of the undervoltage era do not transfer; and the constant-lagging-power-factor load archetypes of planning and protection studies no longer describe the light-load fleet (Section VII)—a qualitative synthesis, not a defence-scheme or load-model redesign, which remain outside the paper's scope. Section III measures the trajectory; Section IV attributes it between fleet and network; Section V identifies the term and tests its scaling; Sections VI and VII set out the overvoltage consequences and the analysis implications; Section VIII concludes with the outlook.

This work extends the author's EEA 2025 conference paper [18], which described the 2009–2024 aggregate trend qualitatively. The present paper adds twelve further years of baseline back to 1997; grid-exit-point resolution; the quantified rate, crossing, and artefact test; the decomposition;

the connection-scaling identification; and the two post-EEA ENTSO-E Final Reports as an independent evidentiary anchor. Consistent with its descriptive scope, the paper reports the measured trajectory and the decomposition without making any policy recommendation.

II. Data and Method

A. Sign convention and primary variable

Reactive power carries a direction that is the entire point of this study. We use the convention $Q < 0$ for *leading* (capacitive, voltage-raising) and $Q > 0$ for *lagging* (inductive, voltage-lowering). We lead throughout with the signed reactive power in MVAr, which is additive across supply points and has no discontinuity at unity, and present power factor only as a familiar comparator. Power factor is non-additive, so where a system power factor is quoted we sum P and Q across the panel first and derive $\mathrm{PF} = \Sigma P/\sqrt{(\Sigma P)^2 + (\Sigma Q)^2}$ rather than averaging power factors.

B. The grid-metering record and the balanced panel

The primary data is half-hourly real and reactive power at every grid exit point (GXP), the substations where the national transmission grid hands electricity to the local distribution networks, from 1997 to 2025, published by the New Zealand Electricity Authority on its Electricity Market Information (EMI) platform. Each per-year file is half-hourly for the whole year (48 trading periods per day), with a real-power and a reactive-power column for each GXP. Over the record, 238 distinct GXPs appear; a balanced core of 132 is present in all 29 years.

We reduce each GXP's year to a small set of features, of which the headline quantities are the average overnight and evening-peak reactive and real power. Following the prior EEA analysis, the overnight window is trading periods 6–10 (approximately 02:30–05:00, the lowest demand) and the evening peak is trading periods 36–38 (approximately 17:30–19:00). We also record, for each GXP-year, the fraction of overnight half-hours that were leading. The full feature table is 5,261 GXP-year rows, one per GXP per year.

The single largest way a 29-year trend could be spurious is a change in the set of measured GXPs over time: new urban, cable-heavy sites appearing and old rural ones dropping out would manufacture a leading trend from bookkeeping alone. The primary defence is the *balanced panel*: only the 132 GXPs present in every one of the 29 years, so any change is a change in the same sites rather than a change in which sites are measured. We call that turnover *churn*; the balanced panel is churn-controlled by construction ("balanced" in the panel-data sense, not the three-phase sense). We follow one rule throughout: *trends on the balanced panel, snapshots on the full panel*. The paper's subject is the distribution networks; restricting the balanced panel to demand networks (excluding a handful of generator, industrial, and rail connection points) leaves 123 GXPs. We report both the 132-point and 123-point figures and keep them labelled; they are the same trend at the same rate (Section III). The residual churn bias is signed: the demand grid exit points excluded from the balanced panel (post-1997 entrants and exits) are by 2025 on average *more* leading than the panel and drift faster, so the balanced panel, if anything, understates the national drift; the extended version [11] carries the churn audit.

C. Metering provenance and accuracy

The active and reactive power at each grid exit point is settlement-grade revenue metering in the highest-accuracy category of the regulatory Metering Code [19]: Class 0.2S for active energy under IEC 62053-22 [20], with the reactive channel held to the looser Class 2–3 of IEC 62053-23 [21]—least accurate exactly at light load and near-unity power factor—and the metered quantity is the fundamental-frequency (*displacement*) reactive power of IEEE Std 1459 [22], the component that drives the voltage rise of Section VI, with harmonic distortion out of scope. The full measurement-error analysis—the meter fleet and its one staggered 2010–2013 replacement, the error classes, the bounds, and the test procedures—is carried in the extended version [11]; what the paper rests on is three results. Random error averages down under aggregation and cannot manufacture a near-monotonic 900-MVAr swing through zero. The one systematic error that does not average down—a phase displacement β in the instrument-transformer chain, injecting a spurious term of about $P \sin \beta$ concentrated in exactly the light-load window this paper measures—is bounded, under a generous common-mode half-degree assumption, at some fifty times smaller than the swing (about 0.1 MVAr/yr of apparent drift); a static displacement of any size falls short of the swing, and a β moving coherently across sites is what the re-base test of Section III-B rejects. And the fleet replacement's possible bias runs in the conservative direction (under-recorded leading reactive power on the older fleet would delay, not advance, the measured crossing): the crossing post-dates the rollout, so the leading regime is observed wholly on the modern fleet, and the per-MW drift within the modern-fleet era alone (2014–2025) is -0.019 MVAr/MW/yr against -0.018 over the full record. We therefore treat the multi-decade *drift* as robust while regarding the standing reactive *level* on any single night as carrying a bounded measurement uncertainty—tens of MVAr at most, a few per cent of the 2025 total—distinct from the composition question of Section IV.

D. The asset (parameter) record

To describe each network's physical make-up we assemble a per-company, per-year table from the Commerce Commission's regulated information disclosures: circuit length split into overhead and underground kilometres at each operating voltage (Schedule 9c, disclosed consistently from 2013), capacitor-bank counts (Schedule 9a), demand and connection counts and embedded generation (Schedule 9e), and the regulated asset base (Schedule 4). The disclosures are mapped to the EMI network codes through a 29-year validated correspondence, time-aware because ownership itself changed under the record; the metering panel is attributed per year from the metering files themselves, so the trajectory follows every change, and the cable-dose analyses of Section IV are run under both a per-year and a physical-continuity (successor-network) attribution, with the sensitivity reported there; the extended version [11] carries the correspondence audit. Two data-quality choices are made explicitly: we retain only the "All"-network roll-up rows (multi-network companies are not double counted) and we flag a 2014–2015 definitional break in the capacitor counts (a widening of reporting scope, not new plant). The capacitor series is counts only, from 2013; we use it as a direction-only proxy.

E. Confounds in the metered reactive power

A grid exit point meter records the *net* reactive power at the boundary, which is not in general a clean distribution offtake:

$$Q_{\text{metered}} = \underbrace{Q_{\text{load}} + Q_{\text{cable}}}_{\text{the subject of this paper}} - Q_{\text{embedded gen}} \pm Q_{\text{TP plant}},$$

where the last two terms are *confounds*: signals mixed into the same metered reading that are not the subject of this paper. An embedded *synchronous* generator (hydro, geothermal, cogeneration) regulates voltage and sets its reactive output by the voltage it sees rather than by local load; and at some substations Transpower voltage-regulating plant—a static var compensator, static synchronous compensator (STATCOM), synchronous condenser, or switched capacitor or reactor bank—shares the substation with the offtake, with the revenue-metering boundary determining whether it registers in the metered flow. One behavioural diagnostic catches all of these, whichever party owns the plant—a network's own large regulating compensation would be flagged identically: a clean distribution offtake's reactive power tracks its load, so a straight-line fit predicting Q from P explains an appreciable share of its variation ($R^2 \approx 0.3$–0.5), whereas voltage-regulated reactive is decoupled from load ($R^2 \to 0$, with a large residual the load does not explain). Computed on three spread-out years (2016, 2019, 2022) so that a flag reflects a stable trait, the screen isolates six decoupled grid exit points, at Christchurch (Islington, Bromley), Auckland (Penrose, Hobson Street), Bunnythorpe, and Tuai; the extended version [11] carries the site-level screen and each site's plant.

This matters overnight specifically: restricting to the overnight minimum removes the *solar* confound—photovoltaics are dark at 03:00—but not the *synchronous*-generation one, since hydro, geothermal, and cogeneration regulate voltage around the clock; the extended version [11] carries a worked public case. We flag these sites rather than discard them silently, and we show in Sections III-B and IV that the trajectory and the decomposition are robust to their removal: the contamination affects the absolute *level*, not the direction, intensity, or attribution of the drift.

F. Analysis methods

The trajectory is summarised by ordinary linear slopes for headline rates and, for robustness, by Theil–Sen slopes [23]: the median of the slopes through every pair of years (406 pairs for a 29-year series), an estimator that a few anomalous years cannot move, used here on both the national and per-GXP series; where the two estimators agree—as they do throughout—a trend rests on no small set of years. The test that the trend is not a metering artefact (Section III-B; workings in the extended version [11]) removes each GXP's own Theil–Sen trend and looks in what remains for a step shared across sites in the same year, using a change-point detector and a coherence statistic. The three decomposition methods are described with their logic and results in Section IV. All analysis is reproduced from public data in the accompanying notebooks.

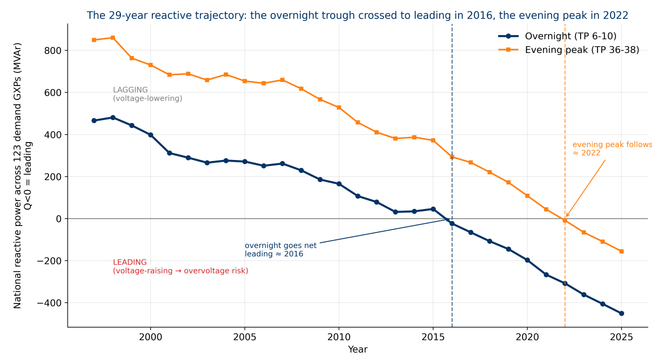

Fig. 1. The 29-year reactive trajectory on the balanced demand panel (123 grid exit points present in all years). Overnight (trading periods 6–10) drifts from strongly lagging ($Q > 0$, voltage-lowering) to clearly leading ($Q < 0$, voltage-raising), crossing net-leading in 2016; the evening peak (trading periods 36–38) follows much later. The diurnal split (the problem lives overnight at low load, not at the busy evening peak) is the signature of a standing capacitive term dominating when load is least able to mask it; Section IV attributes that term mainly to the demand fleet rather than to network charging. Sign convention: $Q < 0$ is leading.

III. The 29-Year Reactive Trajectory

A. Lagging to leading, overnight

Fig. 1 shows the central empirical result. Summed across the balanced demand panel, mean overnight reactive power fell from $+466$ MVAr (lagging) in 1997 to -451 MVAr (leading) in 2025, a near-monotonic linear drift of -31.1 MVAr/yr (robust Theil–Sen slope -31.3 MVAr/yr, 95% confidence interval (CI) $[-34.0, -28.1]$ [23], a rank-based interval indicating spread rather than calibrated coverage; construction in the extended version [11]), crossing into net-leading overnight in 2016. On the wider 132-point balanced panel that also includes non-distribution connection points, the same drift is $+673 \to -296$ MVAr at -30.3 MVAr/yr, crossing in 2020—the same trend, labelled distinctly and never blended into one figure. Expressed per megawatt of overnight load, the drift is about -0.018 MVAr/MW/yr; we lead with this intensity and with the share of networks gone leading because, unlike the absolute MVAr total, they are insensitive to a few large contaminated nodes (Section III-B). The denominator is not doing the work: overnight load on the panel grew from about 1,680 MW in 1997 to about 1,960 MW in 2025. In the familiar comparator of Section II-A, the overnight aggregate power factor moved from 0.964 lagging to 0.975 leading—a numerically small excursion either side of unity concealing a 900-MVAr reversal, which is why the signed reactive power is the working variable throughout.

The diurnal contrast is itself a result. The evening peak crossed net-leading only in 2022 and remains far less leading than overnight, which locates the net-leading regime firmly at low load. Expressed as a fraction rather than a sum, the mean share of overnight half-hours that are leading rose from about 1% in 1997 to about 74% in 2025: overnight leading became routine rather than exceptional.

B. Measured, not an artefact; robust to contaminated sites

A reviewer's first question of any long series is whether the trend reflects the system changing or the measurement changing. Two features of the record answer that directly, and a further test closes the one gap they leave.

First, the drift does not rest on the older, less homogeneous early data. The metering and data pipeline modernised across the record (notably the move to the present Electricity Market Information system), and the one visibly suspect feature is an early step in the overnight leading fraction between 2000 and 2001, consistent with a change in metering vintage. Restricting to the post-2001 era of the present data pipeline removes any such concern, and the drift is if anything slightly steeper: -33.0 MVAr/yr over 2002–2025 against -31.1 MVAr/yr over the full record (the metering-fleet test proper—the 2014–2025 modern-fleet drift—is in Section II-C). A trend manufactured by early-era inhomogeneity would fade on the later data, not strengthen.

Second, a measurement artefact cannot reproduce the trend's physical structure. The net-leading crossing is specific to the overnight window—the evening peak stays lagging for years longer (Fig. 1)—while the deepening beneath the crossings is near-equal in the two windows over 2013–2025, with evening-peak load essentially flat (Section V-A): the signature of a standing capacitive term netted against different amounts of lagging load. The one systematic metering bias that does not average down scales with the metered load ($P \sin \beta$, bounded in Section II-C) and would deepen the heavily loaded evening window far more than the lightly loaded overnight one, which is not observed. Nor is the shift seasonal: winter (June–August) and summer (December–February) months drift at -31.1 and -30.8 MVAr/yr respectively, so the trend is not an artefact of heating-season load composition. The drift also tracks each network's underground-cable intensity (Section IV); a metering error has no reason to correlate with cable kilometres. Nor does a metering artefact at the grid exit point have a mechanism that scales with the customers behind it: the 2013–2025 deepening grows linearly with each grid exit point's connection count, and that connection-scaling term carries the large majority of it (Section V-C). This combination of a diurnally structured, cable-correlated, connection-scaling shift is one that a change in measurement convention cannot counterfeit. The charging calculation of Section IV is not a further witness here: it *partitions* the measured drift rather than reproducing it, so an instrument error would sit inside the residual it assigns to the fleet.

These two arguments address a gradual or staggered change, such as piecemeal meter replacement across sites. A single system-wide re-base is different: a one-date change of sign convention or central processing would step every supply point at once. We test for exactly that, detrending each site by its own robust line and asking whether sites step together in a single year, and find no such event: the change-point candidate years reach coherence of only 67% and 69% against a typical-year 62%, where a genuine re-base would drive coherence toward 100% (procedure and exhibit in the extended version [11]). The lagging-to-leading drift is therefore a measured property of the system, not an artefact of how it was metered.

A second objection, distinct from a metering artefact, is that the trend could be carried by a few grid exit points whose meters do not see a clean distribution offtake (Section II-E). We test it directly by removing them. Dropping the six *decoupled* grid exit points whose reactive power is set by plant or embedded generation (the *clean* cohort), and more strictly also the steady-generation and net-export points (the *clean-strict* cohort), we re-measure the trajectory on the balanced demand panel (the cohort exhibit is in the extended version [11]).

Two quantities behave very differently, and the difference is the point. The absolute total and the MVAr/yr slope *shrink* when the contaminated sites are removed—the 2025 total from -451 to about -300 MVAr and the slope from -31.1 to -23.4 MVAr/yr—because two of the removed nodes (Islington and Bromley) carry large, load-decoupled leading values, so the absolute *level* is partly inflated by contamination. The per-MW drift intensity, by contrast, is essentially unchanged (-0.0175, -0.0178, and -0.0180 MVAr/MW/yr across the full, clean, and clean-strict cohorts), and the share of networks net-leading overnight by 2025 is likewise stable (75%, 74%, 80%). The quantities that describe the phenomenon—its intensity per unit load and its near-universal reach—do not depend on the contaminated nodes; only the headline *size* does. This is why we lead with the per-MW drift and the share-leading migration and treat the absolute spine as a contamination-sensitive companion.

IV. Decomposition: Demand vs Network

The policy-relevant question is how much of the drift is the network's own cables, which inject leading reactive power as a distributed capacitance, and how much is the changing make-up of electricity demand itself: the retreat of inductive plant (simple motors) and the accumulation of power-electronic equipment (LED lighting, switch-mode supplies, heat pumps and other variable-speed drives). We call the second part *organic*: it is in the load fleet, not the wires. If the drift were mostly cables it would be a familiar network-planning problem; if it is mostly the demand fleet it is structural, nationwide, and outside any single network's control. We approach the question three ways; the two statistical methods share a common underground-cable proxy, while the physics-based method fits no cable coefficient and is independent of both. What the demand-side component physically *is*—the question the decomposition leaves open—is taken up in Section V.

A. Method 1: a clean-cohort contrast

History handed us networks at opposite ends of the cable spectrum. Some distribution businesses are almost entirely overhead line and have almost no underground cable; overhead line injects little leading reactive power, so these networks have almost no cable contribution. If their overnight reactive power still drifts leading, that drift can only be the demand fleet. We compare reactive power per MW of load (so network size cancels), load-weighted within each cohort (larger sites count proportionally more), for networks at or below 12% underground (*clean* in this method's sense: almost cable-free, distinct from the decontaminated cohorts of Section III-B) against networks at or above 40% underground (cable-rich); the method exhibits are collected in the extended version [11].

The clean cohort drifts from $+0.373$ to $+0.065$ MVAr/MW (slope -0.0101/yr), ending near balanced; the cable-rich cohort drifts from $+0.205$ to -0.306 MVAr/MW (slope -0.0134/yr), crossing well into leading. The ratio of slopes ($0.0101/0.0134 \approx 0.75$) puts the demand-side share of the cable-rich cohort's drift at about 75%, cable the remaining 25%. The reading rests on two assumptions the split depends on, stated plainly: that the organic per-MW drift is common across the cohorts (to the extent urban load fleets modernise faster than rural ones, the cable share is overstated), and that the at-most-12%-underground cohort carries no material cable term (to the extent it does, the demand share is overstated); the two possible violations pull the split in opposite directions.

The exact ratio also moves with measurement choices—which of Schedule 9c's two circuit-length tables is used (they differ by a few points of percentage underground), the panel definition, and how the grid exit points that changed network (Section II-D) are attributed—spanning the 64–90% collected in Table I, with the clean cohort identical under every choice; each variant, and the effect of removing the two decoupled Christchurch nodes (Section II-E), is worked in the extended version [11].

The *direction* is therefore robust while the *exact ratio* is not, which is why we treat Method 1's ratio as a hypothesis and bring two further methods. And although cable is the minority of the drift, it is the marginal factor that tips a network from balanced into net-leading: the clean cohort ends near balanced while the cable-rich crosses into leading, so cable intensity shapes *where* overvoltage appears first even though demand composition drives the trend.

B. Method 2: a validated dose-response

The second method gives every GXP a single number—the Theil–Sen slope of its overnight reactive power, scaled by its own typical size—and regresses those per-site drifts on the site's cable dose (its network's percentage underground) in a mixed model with a random intercept per company, with uncertainty from a cluster bootstrap that resamples whole companies, because sites under one company share a dose and are not independent. The fitted drift *at zero cable* (b_0) is the organic drift; the slope (b_1) is the extra drift per unit of cable. Validated on synthetic panels with known planted answers before any use on the real data (the protocol is in the accompanying notebooks), the estimator recovers the sign and magnitude of the organic drift reliably, but it point-identifies the organic *share* only when the cable effect is itself identified—and on the real data it is not, so we rest no magnitude on this method; the magnitude falls to Method 3.

On the real data, the organic drift is $b_0 = -0.016$/yr (95% CI $[-0.021, -0.010]$), leading in 100% of bootstrap resamples: unambiguous and dominant. The cable slope is $b_1 = -0.00005$/yr (95% CI $[-0.00037, +0.00012]$), which crosses zero—weakly identified, because cable dose is a company-level trait with only about 23 distinct values—so we deliberately publish no tight interval on the share (at the point estimate it is at least 78%, consistent with 100%; Table I). Three robustness checks hold, each worked in the extended version [11]: adding capacitor intensity to the fit as a control moves b_0 by only about 10% and leaves it strongly leading; the low correlation between cable dose and distributed-generation intensity (-0.22) indicates the organic signal is not an artefact of where embedded generation sits; and restoring the four sites dropped in the Section II-D mapping moves b_0 by under 1%.

A fourth check replaces the dose measure altogether: refitting with each network's computed charging intensity as the dose—Method 3's physical $\omega CV^2\ell$ (Section IV-C) summed and taken per MW of maximum demand—moves b_0 only from -0.0164 to -0.0178/yr, with the cable term still crossing zero. The two dose measures are in fact *negatively* correlated across networks ($r = -0.23$), percentage underground being in part an urban-density measure rather than a charging-intensity one, so two nearly opposite dose definitions returning the same organic conclusion is a stronger robustness statement than either alone (the charging-intensity dose shares its per-MW denominator with the per-site drifts, which can bias the cable slope but not the zero-cable intercept; no reading here rests on the slope under either dose).

Method 1's urbanisation confound applies here too, and in the same direction: the cable dose is a company-level trait, so if urban fleets modernise faster than rural ones some of that difference loads onto b_1 rather than b_0, overstating cable and understating the organic term. The bias is conservative for the reading we take.

C. Method 3: first-principles charging physics

The third method computes the network contribution directly from physics, with no fitted coefficient, so it sidesteps the weak-identification problem entirely. Any energised length of cable or line behaves as a distributed capacitance and injects a fixed leading reactive power $Q = \omega CV^2\ell$, where $\omega = 2\pi \cdot 50$ rad/s, C is capacitance per km, V the operating voltage, and ℓ the length. Underground cable has far more capacitance per km than overhead line. Using the published overhead and underground kilometres at each voltage from Schedule 9c, textbook per-km capacitances with an explicit $\pm35\%$ band, and a band-by-band sum (charging scales with V^2, so high-voltage lines dominate), we compute the physical charging per network per year and subtract it from the measured overnight reactive power; the remainder is organic (the decomposition figure is in the extended version [11]).

The band is set to span published per-km capacitance values across common distribution cable constructions and is carried through to the result rather than quoted and dropped. The organic remainder is a residual, and we name what else it absorbs: transformer magnetising reactive power (lagging and essentially static), series-reactance absorption (minimal at light load), and the distribution capacitor fleet (static-to-shrinking, Section VI-B)—terms that are second-order overnight or bias the residual toward lagging, so reading the residual as the demand-side drift is conservative.

Over 2013–2025 the measured overnight reactive power drifts at -49.9 MVAr/yr across all demand networks, of which the physical charging contributes only -9.2 MVAr/yr (cable length grows slowly) and the organic residual -40.7 MVAr/yr. The organic share of the *drift* is therefore about 80% (physical band 75–88%). Because this method fits no cable coefficient, it is independent of the two statistical methods; the division of labour is collected in Section IV-D. The window is set by the data: the disclosed circuit kilometres it runs on (Section II-D) are recorded consistently only from 2013, so the physics decomposes the 2013–2025 drift, about 45% of the record; extending the $\sim$80% share to the full 29 years is an inference, supported by the direction-consistent statistical methods (which do span the full record) and by the slow growth of cable length, not a second measurement. We stress that this all-network footprint rate (-49.9 MVAr/yr, 2013–2025) is a different dataset from the balanced panels of Section III (-31.1 MVAr/yr on the 123-GXP demand panel; -30.3 on the 132-GXP panel, 1997–2025); these measure different things over different samples and are not interchangeable (a third, separate cable-correlation series of the same public records, reported in the accompanying notebooks [11], differs again).

A separate and more uncertain question is the composition of the leading reactive power *standing on the network today* (the level, as opposed to the drift). On the level the physics points the other way: a large, near-static cable and line charging baseline has always been present, and what changed is that the organic demand shifted from lagging—which used to mask the charging—toward leading, which no longer does: today's net leading reactive power is then mostly long-standing charging *revealed* rather than newly created. The exact level split has a wide physical band and we do not quote a precise share for it; the level is a distinct, weaker claim and must not be confused with the drift result. We carry this only as the qualitative "uncovered charging" reading that informs Sections VI and VII.

D. Triangulation

Table I collects the three methods, which play different roles and are not all independent: Methods 1 and 2 both rest on the same underground-cable proxy, whereas Method 3 fits no cable coefficient at all. All three, though, consume the same Schedule 9c circuit-length disclosures—cohort definition and dose in Methods 1 and 2, the length ℓ in Method 3—so agreement among them cannot diagnose a systematic kilometre error, which would move all three coherently; under-counted cable would push every method toward the demand side.

TABLE I
THREE DECOMPOSITIONS OF THE DRIFT

Method	**Establishes**	**Organic share of drift**
1: clean-cohort contrast (cable proxy)	direction (shares the cable proxy with 2)	$\sim$75% (78% with contaminated nodes removed; 64–90% across attribution, cable-intensity, and panel definitions)
2: dose-response (cable proxy)	direction; dominance at typical doses	$\geq$78% at the point estimate, consistent with 100% (cable term not pinned by the data)
3: charging physics (no fitted cable term)	**magnitude**	$\sim$80% (physical band 75–88%)

The reading follows that division of labour. The *direction* is robust and identified: the organic drift is leading in 100% of bootstrap resamples while the cable contribution to the drift is not statistically distinguishable from zero, and together these imply the drift is dominantly demand-side. The *magnitude* is carried by the physics: the first-principles charging calculation, which fits no cable coefficient, puts the organic share at about 80% (physical band 75–88%). The two statistical methods are directionally consistent but, because the marginal cable effect (the extra drift per unit of cable) is not identified at the few company-level cable values available, do not themselves pin the share. We report the drift as dominantly organic and do not claim a point-identified split. Cable is the minority of the drift but the marginal factor in where net-leading territory is reached first.

A cross-sectional check agrees: the most-leading demand grid exit points are not the most-cabled—they sit at roughly half the cable intensity of the most-cabled group—so cable cannot be what drives the deepest-leading behaviour, independently confirming the demand-side reading (the full cross-sectional typology is in the reproducibility notebooks [11]).

V. The Demand-Side Term Identified: Standing Capacitance Behind Connections

The decomposition attributes the drift to the demand fleet; it does not say what in the fleet is doing it. Two measured properties narrow the answer, a specific physical candidate follows from them, and a scaling test probes the candidate's sharpest prediction.

A. Load-independent, around the clock

First, the drift is load-independent. Over the 2013–2025 window we summarise each grid exit point passing the full contamination screen (the clean-strict criteria of Section III-B, applied here to all sites metered in the window rather than to the 29-year balanced panel) by robust annual medians on wider windows than the Section II features (trading periods 1–12, midnight–06:00, and 35–39, 17:00–19:30). On the 114 such grid exit points with both endpoints metered, the median per-site change in reactive power is -2.25 MVAr overnight and -2.42 MVAr at the evening peak (median per-site ratio 1.06), while evening-peak load was essentially flat (median change $+2.9$%). A change in how the connected load draws reactive power while operating would scale with the load operating—far larger at the peak than at 03:00—and so would the load-proportional metering bias bounded in Section II-C, whose spurious term goes as $P \sin \beta$. A near-equal deepening around the clock is instead the signature of *standing shunt capacitance*: always energised, indifferent to what the load is doing. It also reconciles the diurnal structure of Fig. 1: the two windows drift in parallel, and net-leading surfaces overnight first because the standing component is netted against more lagging load at the peak.

B. A physical candidate: the mandated EMC filter

Second, the capacitance must live behind essentially every connection, because the drift's reach is near-universal across grid exit points (Section III-B). We therefore advance a specific physical candidate, labelled plainly as a hypothesis: the electromagnetic-compatibility (EMC) filter that practically every mains-connected electronic product is required to carry. A switched-mode device chops current at tens to hundreds of kilohertz, and conducted-emission limits (the international CISPR limits [24], a condition of sale in most jurisdictions, New Zealand included, the same obligation reaching household appliances and variable-speed drives through companion standards) oblige its manufacturer to stop that noise at the appliance's own terminals; the universal compliance solution is a passive filter directly across the mains input (Fig. 2).

Its elements carry safety-class names [25] from where they connect: *X-capacitors* bridge line and neutral—the one position where a capacitor is permitted to be large (typically 0.1–2.2 μF), because a short circuit there is an overcurrent event the appliance's fuse clears rather than a shock hazard—while *Y-capacitors* run from the conductors to earth, where a short would electrify the chassis, and are held to nanofarads by touch-current limits; a common-mode choke completes the filter.

At 50 Hz the filter's only material injection is the X-capacitance, which sits across 230 V drawing pure leading current whenever the plug is live—the filter sits ahead of the rectifier and of any standby logic, since the noise it suppresses would otherwise leak even in standby. The arithmetic is $Q = \omega C V^2$, and the fleet is one worldwide design, so the capacitance is chosen once and the mains it lands on sets the draw: the same microfarad that injects 16.6 VAr at 230 V injects 5.4 VAr on North America's 120 V, 60 Hz mains and 3.1–3.8 VAr at Japan's 100 V—roughly three times the standing draw per device here as in North America, and four to five times Japan's. The Y-capacitors are not quite silent where the appliance is earthed, but at nanofarad scale they inject under a per cent of the X-capacitor beside them, so counting X alone biases the estimate very slightly low.

Commercial equipment carries larger input filters: a variable-speed drive holds tens of VAr and upward, energised wherever the installation is left connected. The regulatory structure makes the accumulation one-way: one instrument drives the capacitance into practically every product—the filter is the universal compliance solution, not a named requirement—while, as Section I-A noted, no instrument bounds its fundamental-frequency injection, and a larger X-capacitor is generally the cheaper route to compliance. The X-capacitance's fundamental-frequency current aggregates—tens of devices behind a connection, whole fleets behind a grid exit point—but each device is certified alone, against a standardised artificial mains network, and the sum is, as far as we are aware, bounded at no scale by any instrument; the extended version [11] carries the contrast with the regime's own treatment of accumulating earth leakage, which it bounds at both scales.

Nor is the mandate the newcomer: the conducted-emission requirement has been the sole route to the European market since the start of 1996 [26] and a condition of sale in New Zealand since at least 2001, so the term's youth is the fleet's, not the rule's—what scaled through the 2000s and 2010s is the switched-mode population the rule reaches, while the wire-wound fleet those classes displaced carried no filter at all; the extended version [11] carries the dated regulatory arc, including the conformity levels that single out switched-mode supplies.

The hypothesis is disciplined by what it excludes: the large DC-link capacitor behind the rectifier makes no standing injection (Fig. 2); motor-run capacitors sit in series with the motor's auxiliary winding and are disconnected with it; and rooftop solar inverters contribute almost none overnight (the isolation relay strands the converter-side filter capacitors, leaving only a small grid-side element).

A bottom-up order-of-magnitude estimate—roughly two million households, each with 20–40 permanently connected filter-carrying devices at 2–4 VAr (the population above roughly 25 W, the threshold below which published reference designs make the X-capacitor design-dependent and then absent entirely [11]; roughly 80–320 MVAr in aggregate), plus a commercial and industrial fleet of the same order—puts the national standing total at roughly 150–600 MVAr, or 100–200 VAr per household as the central band.

Sizing its most recognisable member bounds the composition: heat pumps—an adoption jump inside the study window—work out at roughly 37–69 MVAr from census counts and reference-design filter values for the outdoor units [27], [11]. Against the 80–320 MVAr residential envelope that is of order a quarter at central values: a material slice, but a minority, so the bulk of the standing term belongs to the far more numerous small supplies. Because the filter, unlike the heating duty, is energised year-round, the slice is also consistent with the drift's winter–summer invariance (Section III-B); a heating-season mechanism would not be.

The per-device values are physics and datasheet-grade figures, not New Zealand bench measurements—reference designs rather than teardowns of shipped New Zealand product—and the device counts are estimate arithmetic anchored by the intrusive standby surveys [28], [29] rather than a New Zealand census; we grade the attribution accordingly (component-level basis in the extended version [11]). The estimate's value is that it makes two testable predictions: the national scale of the drift, and—because the fleet lives behind consumer connections—that the drift should scale with the number of connections served, with nothing left at zero connections.

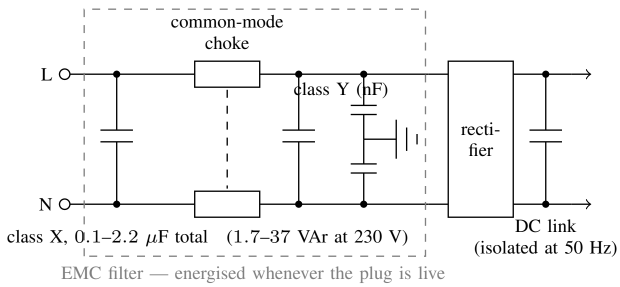

Fig. 2. The mains input stage of a switched-mode device. The EMC filter's X-capacitance sits directly across line and neutral, ahead of the rectifier and of any standby switching, and injects ωCV^2 of leading reactive power whenever the plug is live; the common-mode choke is transparent at 50 Hz, the nanofarad-scale Y-capacitors inject under a per cent of the X-capacitor beside them, and the large DC-link capacitor makes no standing injection, recharging through rectifier conduction pulses only while the device draws power. Component values are datasheet-grade typical ranges, per device rather than per capacitor; on North America's 120 V, 60 Hz mains the same 0.1–2.2 μF range injects 0.5–12 VAr.

The first prediction is met at the order-of-magnitude grade the estimate carries. Summed across the same 114 grid exit points, the median overnight reactive power moved from $+188$ MVAr (net absorbing) in 2013 to -182 MVAr in 2025: a deepening of 369 MVAr; the extended version [11] carries the summary-choice variants. Those exit points serve about 1.5 million connections, near 70% of the national total, so the rise normalises to about 210–280 VAr per connection on the footprint that produced it—above the estimate's 100–200 VAr per household centre rather than inside it. The gap is expected rather than anomalous, and Section V-C quantifies it: a connection is not a household, since the count includes commercial and industrial connections carrying larger filter fleets, and each connection also brings its share of medium-voltage cable. This is a bracket an order of magnitude wide by construction, not an allocation—the estimate bounds today's standing total, of which the twelve-year rise is the recently added part: what it can refute is a bottom-up estimate wrong by that much in either direction, and this one is not.

The dose arithmetic adds a third, cross-country prediction. If the filter is the mechanism, the per-connection drift should order across systems as ωV^2—Japan, then North America, then the 230-volt world—an ordering that cables, heat pumps, and correction plant have no reason to reproduce (Section VIII). It also reframes the transition literature: every system in Section I-A runs 230-volt mains, and we are not aware of an equivalent North American report—on this arithmetic the expected shielded case, a third of the per-device dose spread across interconnections orders of magnitude larger than an island system. The North American record the field does hold confirms the arithmetic rather than contradicting it: the documented North American case of overvoltage instability is Hydro-Québec's "highly capacitive" 735-kV network [12]—extra-high-voltage line charging behind 120-volt outlets, exposure in the network's voltage class rather than the mains-voltage dose. Of the two domestic predictions the second is the more discriminating; Section V-C tests it.

C. A connection-scaling test

Connection counts are public: the retail market-structure dataset on the same regulatory platform records the number of installation control points (ICPs—metered connections) behind each network supply point, monthly, from December 2003 [30]. We sum ICPs across retailers and embedded networks to each grid exit point, take calendar-year means for 2013 and 2025, and join them to the cohort under a documented correspondence audit, since the registry's root-NSP attribution and the metering's exit-point boundaries are not the same partition of the network: 103 of the 114 sites enter (the rest are direct-connect industrial points or lack a registry endpoint), and sites whose registry codes demonstrably swap consumers with no matching movement of metered load are aggregated into station or sibling groups, giving $n = 96$ panel units, with the site dispositions in the extended version [11]. The test regresses each unit's 2013–2025 change in median overnight reactive power on the mean connection count it serves. The decision rule was fixed before the first computation on 6 August 2026, is recorded in the accompanying notebooks, and is applied here unchanged: the hypothesis is supported if the bootstrap 95% confidence interval on the slope excludes zero, the slope point estimate lies within 50–400 VAr per connection, and the fit explains more than 15% of the cross-site variance; a grid-exit-point-level metering artefact, with no mechanism to scale with the connections behind the meter, predicts the mean-only model instead (the same average change at every site).

The result is Fig. 3. The deepening scales linearly with connections served: the slope is -230 VAr per connection, the fit explains 80% of the cross-unit variance, and a through-origin fit agrees (-237). The panel's 96 units serve 1.52 million connections between them and sum to 376 MVAr of deepening, which the fitted line reproduces exactly: 348 MVAr from the slope plus 28 MVAr from the intercept. Section V-B's 369 MVAr is the lower figure because it nets off the sites that shallowed and spans all 114 exit points rather than the 103 that carry a retail registry. The coefficient belongs to this footprint and does not scale to the national connection count, whose connections sit partly behind excluded and contaminated exit points. Because sites under one company share an owner and are not independent, every interval here is reported under two resampling schemes, 4000 replicates each: resampling the 96 units independently gives a 95% interval of $[-272, -199]$, and resampling whole companies—the more demanding test—gives $[-313, -209]$, with a wild cluster bootstrap rejecting the no-scaling null at $p < 0.001$.

All three criteria are met, and the result is not fragile under single-unit or single-company deletion (Fig. 3); the mean-only artefact model loses by an Akaike information criterion difference of about 155, a decisive margin on the standard model-selection score, with the wild cluster bootstrap as its clustered counterpart; and the extended version [11] carries the full robustness battery (rank-based and per-unit estimators, leverage cases, size independence).

The intercept—what is left at zero connections—is -0.29 MVAr per unit and indistinguishable from zero under

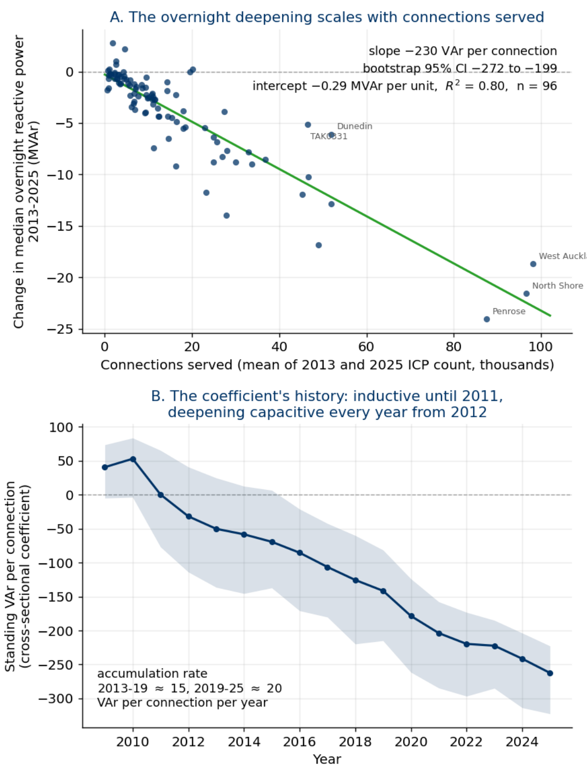

Fig. 3. The connection-scaling test. (A) Each screened panel unit's 2013–2025 change in median overnight reactive power against the mean number of connections (ICPs) it serves ($n = 96$): the change deepens at -230 VAr per connection (95% CI $[-272, -199]$ resampling units, $[-313, -209]$ resampling whole companies) with an intercept indistinguishable from zero (-0.29 MVAr per unit, under a tenth of the cohort deepening once aggregated) and $R^2 = 0.80$ (80% of the cross-unit variance explained); deleting any single unit moves the slope only within $[-242, -220]$, and any single company only within $[-258, -224]$. Nothing about a grid-exit-point metering artefact predicts an effect proportional to the connections behind the meter. (B) The coefficient's history: the per-connection coefficient refitted across units each year on the 92 units present throughout 2009–2025 (shaded: bootstrap 95% CI). On the inductive side of zero until 2011, it crosses in 2012 and deepens every year thereafter at an accelerating rate. Sign convention: negative = added leading (capacitive) reactive power.

every inference scheme (resampling units, resampling whole companies, wild cluster bootstrap $p = 0.29$). Aggregated across the 96 units the point estimate is -28 MVAr, under a tenth of the cohort's 369 MVAr deepening. The bottom-up estimate predicted nothing at zero connections; the measurement meets that prediction on both limbs (the first run of the same pre-specified rule, on the narrower pre-audit 83-site panel, reads the same verdict and is reported unblended in the extended version [11]).

The coefficient also has a history (Fig. 3B). Refitting the across-units regression year by year on the 92 panel units present throughout 2009–2025—each year on the standing level rather than the 2013–2025 change, so the coefficient carries the whole connection-scaling class—the per-connection term starts on the *inductive* side of the axis ($+41$ VAr per connection in 2009, the motor-era fleet's residual signature), crosses zero in 2012, deepens every year thereafter, and reaches -262 VAr per connection by 2025. The crossing year is a point estimate: the yearly intervals straddle zero on both sides of it, and the interval is clear of zero on the capacitive side only from 2016, so the ramp and the post-2016 deepening are what the data establish, not the date. The 2013–2025 movement, -212 VAr per connection, reproduces the two-endpoint slope within 10%—a consistency check across two parameterisations of the same metered panel, not corroboration by an independent route.

The series is smooth, where a staggered metering-replacement artefact would enter as steps, and the in-window accumulation rate rises across the window; the extended version [11] carries the window means. The emergence also overlaps the 2010–2013 meter-fleet replacement (Section II-C), and the coincidence is worth naming: an instrumentation change would enter as a step to a new level, not as a ramp that crosses zero mid-record and is still deepening a decade after the rollout ended, equally in both diurnal windows, and concentrated in the term that scales with connections. The connection-scaling structure, in short, is not a fixture of the network: the fleet's per-connection signature swept from inductive through zero in the early 2010s and is still steepening on the capacitive side. We do not attribute the crossing date to any single device class.

The inductive starting point has a domestic witness: the national household end-use project metered nine houses over 2002–2004 and found the always-on standing load at a power factor of 0.5–0.7 inductive, attributed to motors and "transformer inductive loads" [31]—the iron fleet this record watches leave. Its caveats (nine non-random houses; a total rather than a displacement power factor) travel with it in the extended version [11]: it corroborates the starting point; it does not measure it.

Read honestly, the slope is the whole connection-scaling capacitance class, not the device fleet alone. Each connection also brings its share of medium-voltage cable (low-voltage cable is negligible, since charging scales with V^2), and the registry extract carries no consumer-type field—the counts cannot be split residential from commercial—so commercial connections' larger filter fleets ride in the same coefficient; that the slope sits somewhat above the 100–200 VAr central per-household band is therefore expected rather than anomalous. The units that sit off the fitted line do so in a sensible pattern: grid exit points serving industrial-weighted load (Takanini and Wiri, in south Auckland) deepen less than their connection counts predict, while the Penrose group, serving Auckland's cable-dense inner isthmus, deepens more.

The scaling is also not site size in disguise. Load and connections rise together across sites, so a load-proportional metering error—the $P \sin \beta$ class of Section II-C—would mimic connection scaling; but that class is bounded far below the measured deepening, would concentrate the deepening in the loaded window rather than equally in both (Section V-A), and would over-predict precisely the industrial-weighted sites that in fact fall short of the fitted deepening. What the test establishes at measurement grade is the scaling law itself; the split of the coefficient between device filters and per-connection cable remains hypothesis-grade.

One rival survives in principle: a load-independent metering-registration drift phased in with meter replacement. It has no mechanism to produce the connection-scaling term that carries the deepening, and the connection-independent residual it could sit in is indistinguishable from zero under every inference scheme, so it is heavily disfavoured as an account of the trend while remaining formally excluded only by a measurement that bypasses the grid-exit metering chain. That measurement is identified and simple—night-time reactive flow at a residential distribution transformer with a known house count, where metering on the low-voltage side sees no transformer magnetising and no material cable charging, isolating the device fleet and splitting the coefficient—and is the stated next step.

VI. From Leading Reactive Power to Overvoltage

A. Why leading reactive power raises voltage

The voltage change along a line depends mostly on the reactive power and the reactance, $\Delta V \approx (RP + XQ)/V$, with the XQ term dominant on the grid: lagging reactive power ($Q > 0$, the historical norm) drops voltage toward the load; leading ($Q < 0$, the new overnight reality) pushes it up. Overnight is the dangerous window—demand is lowest, so little load absorbs the reactive power or pulls voltage back, while the cable charging is always on—and the mechanism can run away: charging rises with the square of voltage, and a protection trip that disconnects a transformer or load leaves the remaining network carrying more charging per unit of load. Two distinct feedback loops run through that description, usually described as one. In the first, the *demand-side* loop, disconnected load takes with it whatever reactive absorption it was providing, and voltage rises; whether shedding load raises voltage or lowers it therefore turns on the power factor of what comes off, since lagging load removes absorption while sufficiently leading load removes a reactive source instead. In the second, the *plant-side* loop, rising voltage disconnects plant and generation that were absorbing, and voltage rises further; that loop runs on the response of protection to voltage and would run the same way behind a demand of any power factor. In isolated and autonomous systems the feedback is well documented [32], [33]; in neither 2025 blackout below did the lost absorption leave with the demand (Section VI-C).

B. The system is already responding

The New Zealand system is visibly responding from both ends. Distribution capacitors are coming out: from the Commerce Commission disclosures, the national bank count fell from 298 in 2015 to 254 in 2025 (counts, not MVAr)—a static-to-shrinking capacitor fleet cannot be causing a rising leading trend, and the removals are the signature of the opposite, high-voltage problem. Transmission absorption is going in: shunt reactors and STATCOMs installed against operator security forecasts that record the practice of managing overnight high voltages by switching selected transmission circuits out of service [34], with the follow-up study finding no significant overvoltage issues for current and future light-load conditions once the reactors and a STATCOM were in service [35]; the extended version [11] carries the asset list and ratings. And the operator has named the demand-side term: Transpower's 2023 transmission planning report expects the high-voltage issue to worsen and lists "load characteristics becoming generally more capacitive" among the factors [36]—the asset owner, in a planning document, attributing part of the standing pressure to the electrical character of the load, the quantity this paper measures.

C. The realised risk, abroad

The regime is not hypothetical. The Iberian event of 28 April 2025 was found to be a voltage and reactive collapse—the Final Report identifies "non-effectiveness of voltage control" as a root cause, and the absorption the cascade removed left with the generation, tripped on overvoltage protection, not with the demand [16]. The North Macedonia event of 18 May 2025 was a cascade of transformer trips from sustained overnight overvoltage, driven by the high capacitive reactive power of light-load networks with no compensation plant on the system and no load shed at all, so it ran with the demand-side loop never operating [17]. Each event had local specifics; we read them as corroboration that the leading-reactive regime is a realised blackout mechanism, with New Zealand's distinctive contribution the 29-year measurement and decomposition of how a system drifts into that regime in the first place; the extended version [11] carries the event detail.

D. Coverage of existing reactive-power provisions

New Zealand's reactive-power rules were written for the old problem of lagging load pulling voltage down at the daytime peak: the connection minimum power factor is a term of the default transmission agreement that binds the distributor at the connection point—the right party, at the boundary where Section III takes its readings—but it applies during regional peak demand periods and at no other time, the periods themselves fixed by reference to a transmission pricing methodology replaced in 2023 [37]; the reactive-demand measurement provision operates only on weekday hours, 07:00–21:00 [38]; and published power-factor charges run on a lagging basis (e.g., [39]). The one window where overvoltage now lives is the window the rules do not cover. The rules did not *cause* the leading surplus—Sections IV and V place the driver in the product-standard demand fleet, outside their reach—but neither do they cover, price, or limit it; whether that should change is a policy question outside this paper's descriptive scope (the provision detail is in the extended version [11]).

One rule moved as the record closed: on 13 November 2025 the statutory low-voltage supply band was widened from 230 V $\pm 6\%$ to $\pm 10\%$ [40], so about seven weeks of the 2025 endpoint sit after commencement—small, and running with the reported drift rather than against it, but the measurement is not strictly pre-commencement and we do not claim it as such. Standing capacitance draws $Q = \omega C V^2$, so the same fleet delivers about 2% more reactive power for each 1% of sustained voltage elevation the wider band accommodates; post-2025 updates of the per-connection coefficient should control for measured voltage or name the confound (the ceiling arithmetic is in the extended version [11]).

VII. IMPLICATIONS FOR POWER-SYSTEM ANALYSIS

The measured reversal relocates the binding voltage-security question. For the lagging era's failure mode—voltage collapse under heavy load—power-system practice holds a mature, layered stack: automatic under-voltage load shedding as a last resort, tap-changer blocking, PV- and QV-margin analysis, and decades of operational tooling, all built on load models of an inductive fleet. In the overnight regime measured here the equivalent layers are thinner: the *automatic* last-resort layer is addressed to the opposite direction, and the margin tooling has no counterpart of comparable maturity.

The standard emergency lever was built for the falling direction; in this regime its effect on voltage is no longer of one sign. In a collapse, shedding load is corrective; in the light-load capacitive regime the two loops of Section VI-A separate. What shedding does to voltage in the demand-side loop turns on the power factor of what comes off—lagging load takes absorption away with it; sufficiently leading load takes a reactive source away instead—and that power factor is the quantity Section III measures moving through zero, so a scheme designed on the lagging assumption can no longer assume it. The plant-side loop is indifferent to what the demand draws—protection that removes absorbing plant raises voltage behind a demand of any power factor—and it is the loop the Iberian cascade ran on; in North Macedonia the demand-side loop never operated at all (Section VI-C).

The Rhodes cascade of 2016 is the one investigated event in which the demand-side loop can be seen operating: under-frequency shedding raised voltage and deepened generator reactive absorption—an aggravator in a collapse ultimately triggered on loss-of-field, and one a small shunt reactor would have prevented [32]. No investigated event attributes a blackout to the demand-side loop. And on the demand character measured here the implication runs further: net-leading demand takes a reactive source with it when it trips, so the classic load-rejection voltage rise weakens and can reverse in sign, and the overvoltage exposure in an event concentrates on the plant side—the absorbing plant and generation that protection removes, and the under-excited operating points where the autonomous-system record already places the failure path and its countermeasures [32]. The last-resort schemes discussed here were all designed for the other direction: none connects load or absorption when voltage runs away, and none reads the power factor of what it sheds.

The driver resists dispatch, and the cascade reinforces itself. The *drift* into the regime is dominantly the demand fleet, but the standing overnight surplus the mechanism feeds on is substantially long-standing network charging the fleet's shift has uncovered (Section IV)—distributed, always energised, and not itself dispatchable short of spending N-1 security on voltage margin (Section VI-B). And the cascade arithmetic is inverted relative to collapse: each element lost in a voltage collapse reduces the demand that is depressing voltage, so the disturbance partly self-relieves, whereas each definite-time protection operation in an overvoltage cascade removes absorption and strengthens the driver—self-reinforcing in quantised, irreversible steps, running in well under a minute in Iberia [16] and over hours in North Macedonia [17].

The analysis stack under-represents the regime it is entering. Planning and protection studies still commonly represent demand with constant, lagging power-factor archetypes calibrated to the motor-dominated fleet [7], [8]; the measured net overnight flow now carries the opposite sign. And while overvoltage protection and locally automatic shunt switching exist as device-level controls, and the QV curve's upper limb, temporary-overvoltage studies and self-excitation screening each address part of the question, none is reported as a routine margin quantity the way PV and QV margins report distance to collapse; the field's 2026 mechanisms review likewise proposes containment countermeasures but no margin quantity for the upper band [12].

The measurement gives the load-model correction a concrete, portable form. The light-load fleet needs a standing shunt-capacitance term that scales with connections served—on the New Zealand panel, about 220 VAr per connection accumulated over 2013–2025 (Section V-C; a boundary-level figure that carries each connection's share of network cable with it)—alongside whatever power-factor archetype represents the operating draw. A constant-power-factor archetype cannot represent this term even qualitatively: it is present when the load is not, it accumulates behind existing connections rather than arriving with demand growth, and it is young and moving (Section V-C)—archetypes calibrated even a decade ago predate it entirely.

None of this makes the regime unmanageable. With fast absorption installed, New Zealand's operator reports no significant overvoltage issues under current and future light-load conditions (Section VI-B): the gap is not feasibility but architecture—the automatic defences, the security-margin tooling, and the load models were built for the direction the system has left. Their redesign is development work beyond this paper's descriptive scope. What the measurement supplies is the direction and the rate: a national boundary flow that has crossed to the capacitive side of the axis around which those tools were built, driven dominantly by the load fleet, at about -0.018 MVAr per MW of overnight load per year, with the crossing already behind it.

VIII. CONCLUSION

Using 29 years of regulatory half-hourly metering, we measured a national power system's reactive character shifting from lagging to leading at supply-point resolution: on the balanced demand panel (the same 123 grid exit points throughout), mean overnight reactive power fell from about $+466$ MVAr to about -451 MVAr (about -31 MVAr/yr), crossing net-leading overnight in the mid-2010s and at the evening peak only later—a measured phenomenon, not a metering artefact, holding in winter and summer alike, within the modern metering-fleet era alone, and with every contaminated grid exit point removed. The drift is dominantly the demand side: a first-principles charging calculation that fits no cable coefficient puts the organic share at about 80% (physical band 75–88%) over the 2013–2025 asset-disclosure window, with the two statistical methods corroborating the direction over the full record but not pinning the split. And the connection-scaling test identifies what the demand side is physically doing: at about 220 VAr per connection added over 2013–2025, with no detectable connection-independent residual, from a term that sat on the inductive side of zero early in the 2010s and has accumulated faster over the second half of the window than the first, the load fleet is not merely drawing fewer lagging vars overnight; it is accumulating standing distributed capacitance of its own.

The significance of the reversal is what it does to the analysis. The defences, margins, and load models of power-system practice were built for lagging load and the undervoltage direction; the measured overnight flow has crossed to the other side, where no last-resort scheme is addressed to voltage rising and each protection operation reinforces the disturbance (Section VII); the same regime has produced two realised blackouts abroad, and the New Zealand system is already responding to it from both ends while its deeper demand-side driver remains unnamed in the rules. New Zealand's record—an advanced-state, small, isolated, heavily cabled analogue whose net-leading crossing preceded the 2025 European events by nearly a decade—is, to our knowledge, the longest direct measurement of that crossing, and both records the identification rests on are ones any operator already holds: if the standing term is carried by the globally standardised product fleet, every system connected to the same supply chain is accumulating it on a similar schedule, at a dose set by its supply voltage (Section V-B)—on 230-volt systems roughly three times the North-American rate per device and four to five times the Japanese, so the dose ordering maps the exposure geographically and is itself the decisive comparative test. Supply voltage sets the dose; transmission voltage sets its visibility. The term is created at the mains, indifferent to the network above it, while the boundary flow that would reveal it sits beneath the network's own charging, which grows with the square of transmission voltage and with route length—so New Zealand's 220-kV maximum is an unusually transparent window, and in a 400-kV system the same term sits beneath a network term several times larger, where a network-side reading of the surplus is the natural one (the reading both 2025 investigations took, Section I-B). The claim is therefore priority of measurement, not of exposure: the failures arrived first where the network term is largest, not where the demand-side term is clearest. The systems with the most to gain from such a record are the ones without one: larger fleets, no measurement of the demand-side term, and, in Europe's case, a demonstrated failure mode.

The record also says where this goes. The drift shows no sign of slowing: the per-connection coefficient accumulated faster over the second half of the window than the first, was running at about 30 VAr per connection per year at the 2025 end, and the next device waves—an electric-vehicle charging and home-battery fleet expected to carry the largest input filters yet—are still arriving; nothing in the record yet marks the turn. Tracking that coefficient with a renewal model of device generations, and the decisive attribution measurement at a residential distribution transformer (Section V-C), are the nearest work (both detailed in the extended version [11]).

The stability question has moved with the sign: with overnight demand now net leading, defence design points at plant-side absorption—ride-through of absorbing plant, coordination of under-excitation limits on generators feeding capacitive networks [32]—and at solved network cases of the light-load state, which this boundary record cannot settle; and because the same V^2 scaling sets a network's own charging per kilometre, systems pairing extra-high-voltage networks with 230-volt fleets—Iberia's 400 kV among them—carry both terms at once. The prediction also runs the other way: unlike the passive filter fleet, the arriving wave of vehicle chargers, photovoltaic and battery inverters is controllable, and interconnection standards already define the voltage–reactive-power response modes to steer it [41], so large inverters operated, or required, to absorb at light load—running in the lagging quadrant against the fleet they share a board with—are the natural first counter to the leading surplus at the grid exit point. Nearer the customer, the distribution networks the reverse flow transits hold two margins in records they already keep: tap-changer position histories—how much regulation headroom remains at light load—and the loss cost of the standing circulating current.

The limitations are stated throughout: the balanced panel controls churn but does not eliminate it; the decomposition attributes the drift, not a causal mechanism in any single load; the exact organic share and the level composition are not point-identified; the device-level attribution of the connection-scaling capacitance remains a labelled hypothesis, with its decisive low-voltage measurement identified; and a single national system is one observation—offset, not cured, by its position as a leading-edge analogue and by the two independent European corroborations. The defence-scheme and load-model development this measurement motivates is separate work. What this paper offers is the measured trajectory, its decomposition, the term it identifies, and the direction power-system analysis must now build for.

Reproducibility

Every number and figure in this paper is computed from public data (Electricity Authority grid metering and Commerce Commission information disclosures) in an accompanying set of analysis notebooks, which also document the path from the raw public records. An extended version of this paper, with the complete measurement-error analysis and the supporting method exhibits, accompanies the notebooks in the reproducibility repository [11]. A companion study, in preparation, extends the record into a fitted circuit model of every grid exit point—a standing term with drift and dated operational steps, load-composition and series terms, and embedded-plant signatures—each element validated against its own natural experiment. Release of any underlying extract is subject to the data custodian's terms.

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